

HW10

Acknowledgement: For Q1 & Q2 I did search online for MATLAB, particularly plotting

Q1:

A:

Transfers:

H_butter =

1.559e12

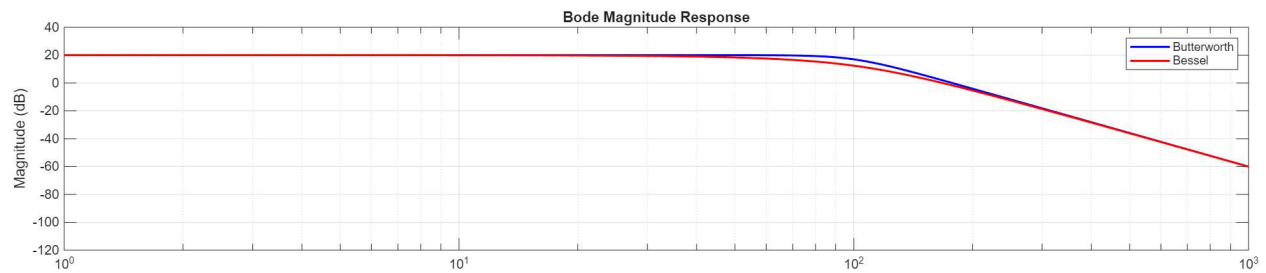
$$s^4 + 1642 s^3 + 1.348e06 s^2 + 6.482e08 s + 1.559e11$$

H_bessel =

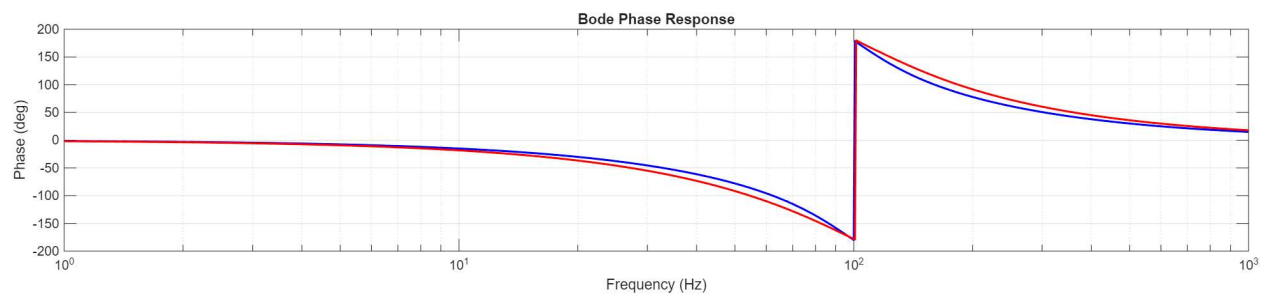
1.559e12

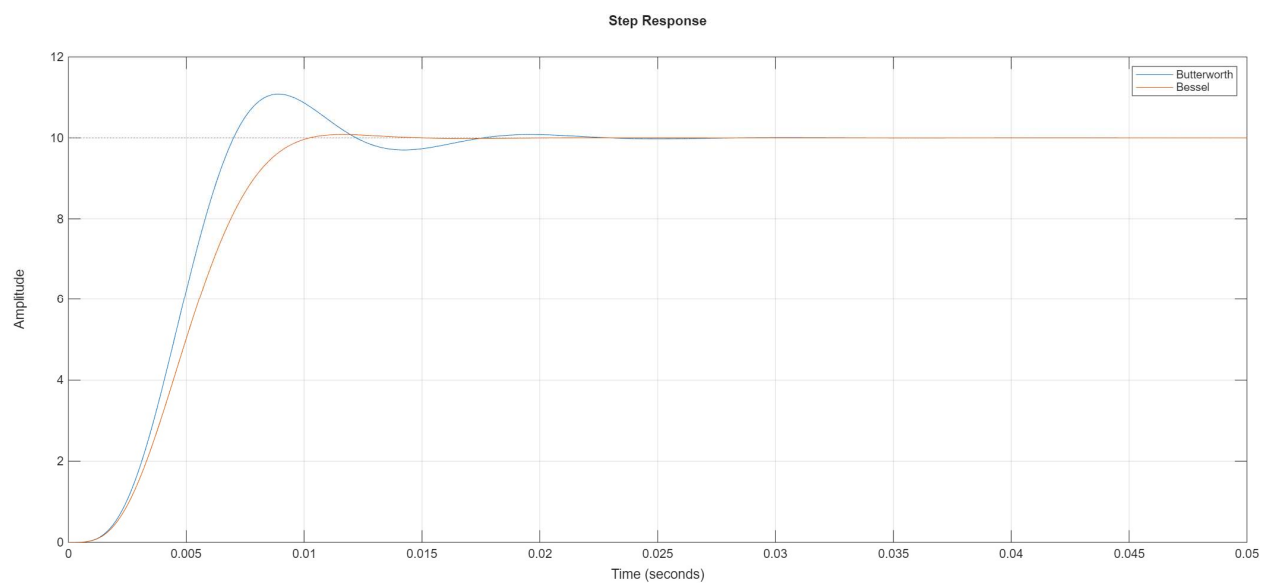
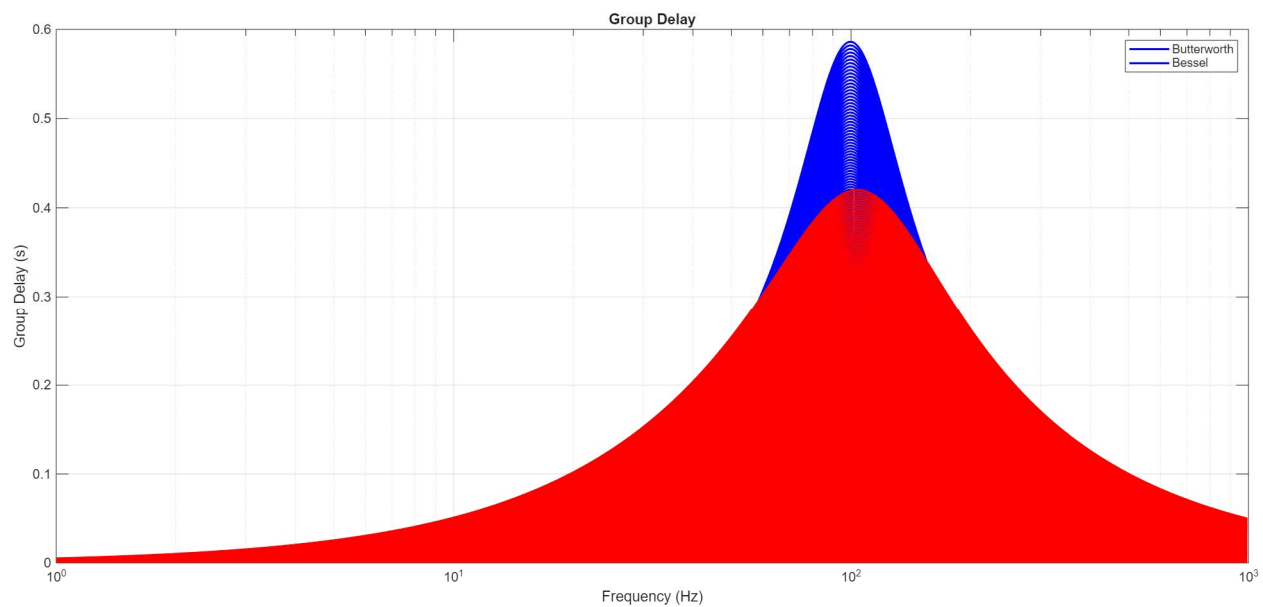
$$s^4 + 1963 s^3 + 1.734e06 s^2 + 7.94e08 s + 1.559e11$$

Bode Magnitude:



Bode Phase:





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clc; clear all; close all;

% Filter specifications
n = 4;           % Filter order
f_c = 100;       % Cutoff frequency in Hz
K = 10;          % Gain
w_c = 2*pi*f_c; % Convert Hz to rad/s
% Butterworth filter (Analog)
[bb, ab] = butter(n, w_c, 's');
H_butter = tf(K*bb, ab);
% Bessel filter (Analog)
[bb_bes, ab_bes] = besself(n, w_c);
H_bessel = tf(K*bb_bes, ab_bes);
% a. Display transfer functions
print('Butterworth Transfer Function (K=10):')
H_butter
print('Bessel Transfer Function (K=10):')
H_bessel
% Frequency vector for plots
f = logspace(0, 3, 1000); % 1 Hz to 1000 Hz
w = 2*pi*f;
% Calculate Responses
H_butter_resp = squeeze(freqresp(H_butter, w));
H_bessel_resp = squeeze(freqresp(H_bessel, w));
% b. Bode plots
figure;
subplot(2,1,1)
semilogx(f, 20*log10(abs(H_butter_resp)), 'b'); hold on;
semilogx(f, 20*log10(abs(H_bessel_resp)), 'r');
grid on;
ylim([-120 40]);
xlim([1 1000]);
ylabel('Magnitude (dB)'); title('Bode Magnitude Response');
legend('Butterworth', 'Bessel');
subplot(2,1,2)
semilogx(f, angle(H_butter_resp)*180/pi, 'b'); hold on;
semilogx(f, angle(H_bessel_resp)*180/pi, 'r');
grid on; ylabel('Phase (deg)'); xlabel('Frequency (Hz)');
title('Bode Phase Response');
% c. Group delay calculation
% Group Delay = -d(phase)/dw
gd_butter = -diff unwrap(angle(H_butter_resp)) ./ diff(w);
gd_bessel = -diff unwrap(angle(H_bessel_resp)) ./ diff(w);
figure;
semilogx(f(1:end-1), gd_butter, 'b'); hold on;
semilogx(f(1:end-1), gd_bessel, 'r');
grid on; xlabel('Frequency (Hz)');
ylabel('Group Delay (s)');
title('Group Delay');
legend('Butterworth', 'Bessel');
% d. Step response
t = 0:0.0001:0.05;
figure;
step(H_butter, t); hold on;
step(H_bessel, t);
grid on; title('Step Response');
legend('Butterworth', 'Bessel');

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Q2:

Transfer:

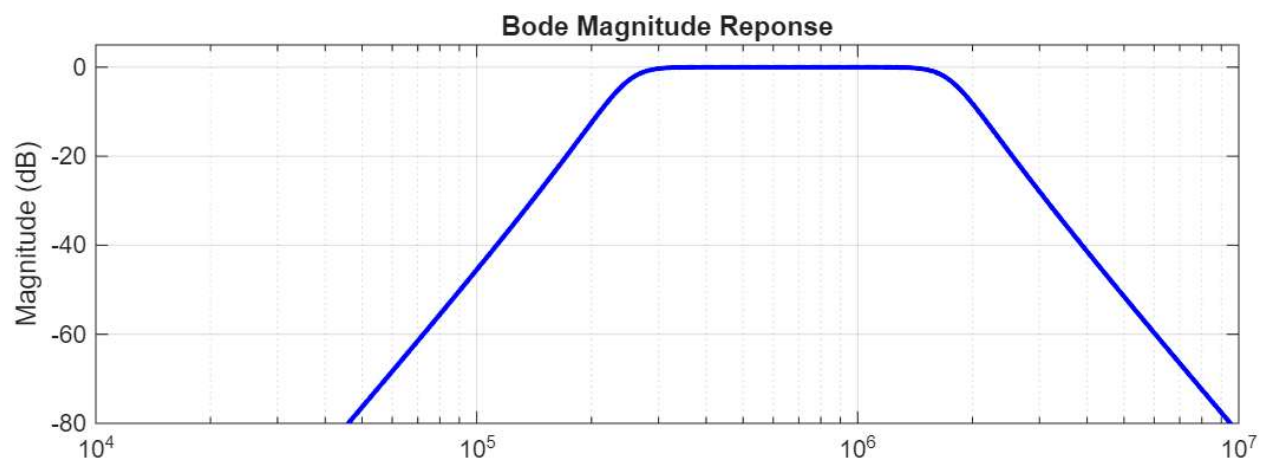
$H_{\text{bpf}} =$

$$7.436e34 s^5$$

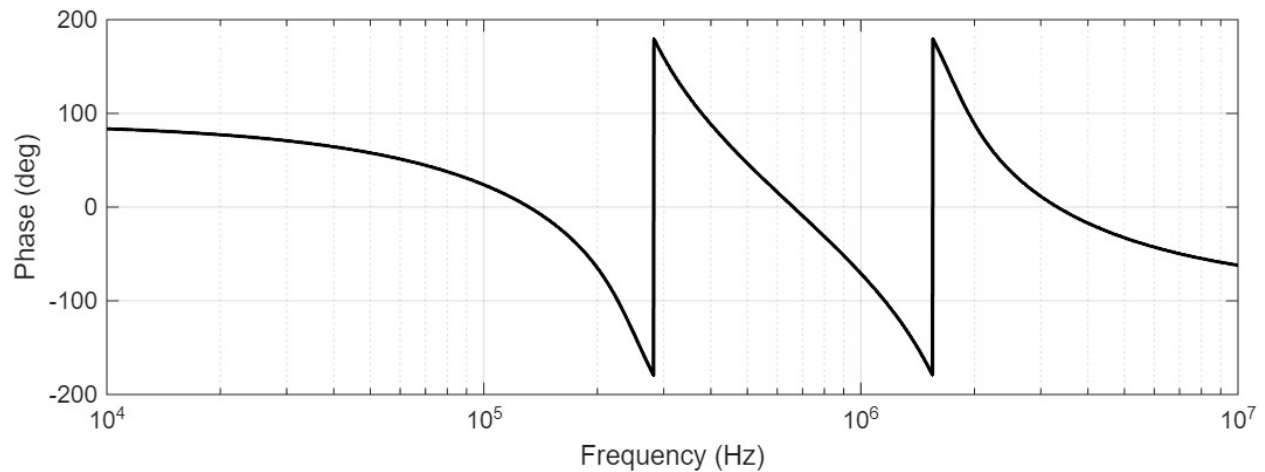
$$s^{10} + 3.05e07 s^9 + 5.515e14 s^8 + 6.491e21 s^7 + 5.262e28 s^6 + 2.804e35 s^5 + 9.088e41 s^4$$

$$+ 1.936e48 s^3 + 2.841e54 s^2 + 2.714e60 s + 1.537e66$$

Bode Magnitude:



Bode Phase:



Upper Cutoff (fc2): 1750.00 kHz

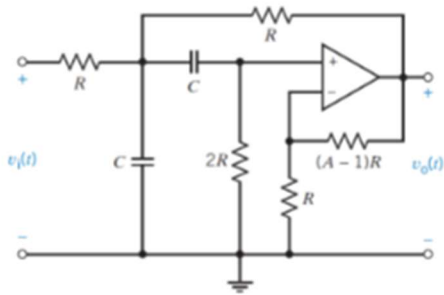
Center Frequency (fo): 661.44 kHz

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clc; clear; close all;
% 1. Filter Specifications
n = 5; % Filter order (prototype)
f_c1 = 250e3; % Lower cutoff frequency (Hz)
beta = 1.5e6; % Bandwidth (Hz)
K = 1; % Passband Gain
% Calculate upper cutoff and center frequencies
f_c2 = f_c1 + beta; % Upper cutoff
f_center = sqrt(f_c1 * f_c2); % Geometric center frequency
% Convert frequencies to rad/s for analog design
w_band = 2*pi*[f_c1, f_c2];
% 2. Design Butterworth Bandpass Filter
% Note: An n-th order BPF results in a 2*n order transfer function
[num, den] = butter(n, w_band, 's');
H_bpf = tf(K * num, den);
% 3. Frequency Response Calculation
f = logspace(4, 7, 2000); % 10 kHz to 10 MHz
w = 2*pi*f;
H_resp = squeeze(freqresp(H_bpf, w));
% 4. a. Display Results
fprintf('--- Filter Analysis Results ---\n');
fprintf('Lower Cutoff (fc1): %8.2f kHz\n', f_c1/1e3);
fprintf('Upper Cutoff (fc2): %8.2f kHz\n', f_c2/1e3);
fprintf('Bandwidth (beta): %8.2f kHz\n', beta/1e3);
fprintf('Center Frequency (fo): %8.2f kHz\n', f_center/1e3);
fprintf('\nTransfer Function:\n');
H_bpf
% 5. b. Plotting the Bode Plot
figure('Color', 'w', 'Position', [100, 100, 800, 600]);
% Magnitude Response
subplot(2,1,1);
semilogx(f, 20*log10(abs(H_resp)), 'b', 'LineWidth', 2);
grid on; hold on;
ylabel('Magnitude (dB)');
title(['Bode Magnitude Response']);
xlim([1e4 1e7]); ylim([-80 5]);
% Phase Response
subplot(2,1,2);
semilogx(f, angle(H_resp)*180/pi, 'k', 'LineWidth', 1.5);
grid on;
ylabel('Phase (deg)');
xlabel('Frequency (Hz)');
xlim([1e4 1e7]);

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Q3:



A: Bandpass, either shorts to ground or blocks input at extrema.

B: $V_+ = V_- = V_o * (R/(A-1)R) = V_o * R/RA = V_o/A$

$$(V_1 - V_{in})/R + (V_1 - V_o)/R + V_1 sC + (V_1 - V_o/A)sC = 0$$

$$V_1/R - V_{in}/R + V_1/R - V_o/R + V_1 sC + V_1 sC - V_o sC/A = 0$$

$$V_1(2/R + 2sC) = V_{in}/R + V_o/R + V_o sC/A$$

$$V_1(2/R + 2sC) = (V_{in} + V_o)/R + V_o sC/A$$

$$2V_1(1+sCR/R) = (A(V_{in} + V_o) + V_o sCR)/RA$$

$$V_1 = \frac{A(V_{in} + V_o) + sCRV_o}{2A(1+sCR)}$$